LAKSH BHARANI

New Grad Software Engineer | Virginia Tech CS + Math & AI | May 2027

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EDUCATION

Virginia Tech, *B.S. in Computer Science* Aug 2024 – May 2027 | Blacksburg, VA
Minors - Mathematics, Artificial Intelligence
GPA: 3.81/4.0
Honors: Dean's List (All Semesters)
Core Courses - Data Structures & Algorithms, Software Design, Computer Organization & Architecture, Discrete Math, Cloud Computing

SKILLS

Languages: Python, JavaScript, TypeScript, Java, Go, C, C++ • **Backend & Cloud:** Node.js, Firebase, Google Cloud Platform, Azure, Docker, PostgreSQL, Linux • **Frontend:** React, Tailwind, SwiftUI • **Concepts:** Distributed Systems, Networking, Microservices, CI/CD

WORK EXPERIENCE

Microsoft, *Software Engineering Intern* May 2026 – Aug 2026 | Redmond, WA
- Shipped an ADA-compliant React/.NET QC platform for **16K+** contracts, saving **10K+** review hours and **$500K/year** in vendor spend.
- Built an agentic workflow platform on **MAF + Azure AI Foundry**, accelerating iteration by **90%** and reducing extraction errors by **20%.**
- Architected a backend-for-frontend layer decoupling clients from **3** independently deployable containerized microservices.
- Enabled secure and independent **EV2** rollouts and **Azure Container Apps** scaling with **Entra ID**, **RBAC**, and data-plane authorization.

Shoonya Life, *Software Engineering Intern* May 2025 – Aug 2025 | San Francisco, CA
- Shipped an AI-based Slack wellness assistant with journaling, program discovery, Google Calendar integration, and guided exercises.
- Cut **p90** inference cost **79%** ($3.61 → $0.75) and token usage **90%** by optimizing chunking, embeddings, and vector-index retrieval.
- Reduced **p50** latency **16%** by introducing asynchronous retrieval and improving the LLM inference pipeline. **Blog Link**
- Deployed to production on **GCP** & **Slack Marketplace**, supporting enterprise workflows through a production-grade Slack integration.

PeakMind.in, *Software Engineering Intern* Jun 2024 – Jul 2024 | Bangalore, India
- Reduced FCP by **57%** (3.5s → 1.5s) in a **Flutter** iOS app through query optimization, caching, and asynchronous processing.
- Improved Speed Index by **44%** (5.2s → 2.9s), profiling and validating performance optimizations with **Google Lighthouse**.

PROJECTS

L4 TCP Load Balancer, *Go · Docker · Git*
- Built an L4 TCP load balancer in Go with round-robin, IP-hash, least-connections, and adaptive load balancing.
- Sustained **84.5K req/s** at **6.7ms p99** across **10K** concurrent connections with zero failures in load testing. **GitHub**

InfraMapper - Agentic Cloud Infra Solution, *Next.js · FastAPI · Azure AI · Semantic Kernel*
- Built a multi-LLM **Azure** infrastructure agent that converts ambiguous user intents into executable cloud deployment plans.
- Implemented critic/self-review loops and post-execution verification to detect unsafe or incorrect resource changes. **GitHub**

nux - AI Powered Native macOS Terminal, *SwiftUI · AppKit · Gemini API · OpenAI API*
- Built a native **macOS** terminal with **SwiftUI/AppKit**, integrating **Gemini** and **OpenAI** for command reasoning and error remediation.
- Designed session-based **state management** and natural language processing for terminal workflow automation. **GitHub**

CLUBS & LEADERSHIP

E-Nable, *Software Team Lead*
- Led software development across a **10**-person team supporting tachycardia detection research and ML model training.
- Built real-time cardiac data collection with iOS-Arduino integration for sensor-driven training workflows.

VT IDPro, *Software Team Lead*
- Led software development for a **4**-person campus navigation kiosk team serving **30K+** annual campus visitors.
- Delivered the end-to-end software platform while coordinating integration with hardware, display, and CAD workstreams.

RESEARCH

AbsInf: A Lightweight Object to Represent float('inf') in Dijkstra's Algorithm, *arXiv* Apr 05, 2025
- Developed a lightweight float('inf') replacement for **Dijkstra's** algorithm, improving runtime by up to **17.2%**.
- Validated performance gains across diverse graph datasets with Welch's t-test ($p < 0.01$). **arXiv**